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Enhancing Drilling Performance Through AI-Based High-Speed Intelligent Telemetry

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Abstract

Traditionally, drilling performance improvements have centered on optimizing physical tools such as drill bits, motors, and rotary steerable systems (RSS). However, challenges in real-time data acquisition persisted, particularly in complex and extended-reach wells. To address these challenges, an advanced telemetry system has been deployed on some drilling rigs, achieving groundbreaking advancements in operational efficiency, cost management, and environmental sustainability.

The technology is designed to support complex drilling bottom hole assemblies (BHAs) equipped with advanced Logging-While-Drilling (LWD) tools. By leveraging artificial intelligence (AI) and advanced algorithms, the system improves signal processing from mud pulses, eliminating drilling and surface noise to enable higher bandwidth and faster data transmission. This high-fidelity, real-time downhole data empowers precise geosteering, providing deeper insights into subsurface conditions. These capabilities allowed streamlining decision-making during drilling operations, significantly enhancing performance in challenging drilling environments.

The results were transformative wherein drilling time for geo-steering lateral sections utilizing deep and ultra-deep resistivity LWD BHA were reduced by up to 50%, resulting in substantial time and cost savings. The enhanced efficiency also had a positive environmental impact by lowering carbon emissions and optimizing resource usage.

The adoption of this new AI-Based high telemetry technology marks a pivotal shift in drilling operation, setting new benchmarks for technological innovation, performance optimization. By addressing key challenges in real-time data acquisition and geo-steering precision, this deployment exemplifies how advanced telemetry systems can redefine drilling operations in today's energy landscape.

Introduction, Field Background and Operational Challenges

Some areas historically drilling practices due to real-time data transmission limitations and time-consuming connection procedures. An integrated engineering approach was deployed in the latest wells, combining high-speed telemetry, dynamic continuous surveying, and a real-time connection optimization strategy. This allowed drilling to continue seamlessly after connections without survey hold, and enabled high-quality

LWD imaging even at high rates of penetration (ROP). The achieved average ROP increased by 130% compared to similar wells field average, while connection times were reduced by more than 50%. Thus, total 8.5-in lateral sections duration was reduced by 60%. This paper outlines the engineering design, real-time decision workflow, and field performance across multiple wells.

The deployment was in lateral sections of a developed field with complex geological environment where drilling operational practices were mainly centered around LWD data quality and its resolution for right well placement rather than speed. A major contributor to reduced drilling efficiency was the limited telemetry bandwidth available with conventional systems, which restricted the ability to transmit high-resolution Deep and Ultra Deep Resistivity image data from LWD tools in real time. While other LWD curves remained unaffected, real-time image quality degraded significantly at higher ROPs. As a result, ROP was often limited to accommodate telemetry constraints.

Stationary surveys have long been a reliable method for confirming well trajectory and enabling informed geosteering decisions. However, maintaining the BHA in a static position in open hole—particularly in extended-reach or horizontal sections—can increase the risk of mechanical issues. Without continuous rotation or fluid circulation, drilled cuttings tend to settle along the low side of the wellbore, forming cuttings beds that elevate torque and drag or, in some cases, contribute to pack-off events (Figure 1). The adoption of dynamic survey techniques reduced the duration of these stationary periods, thereby minimizing associated risks while maintaining survey quality.

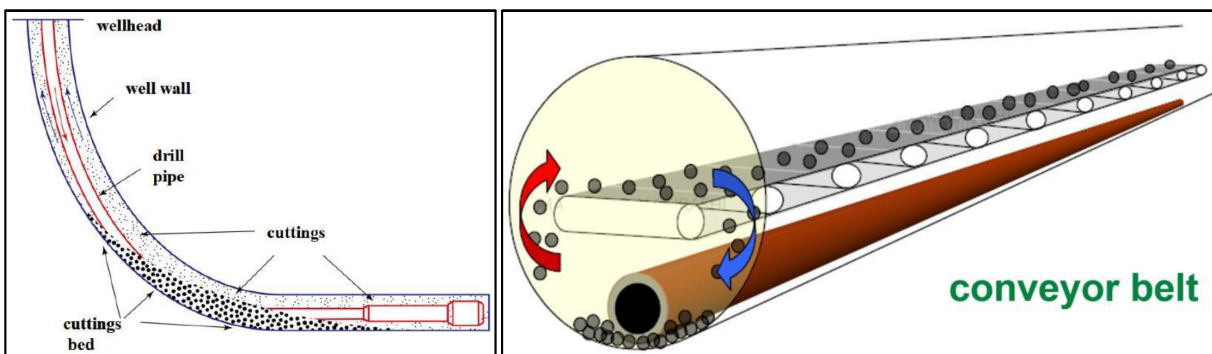


Figure 1—Illustration of cuttings bed formation during stationary BHA conditions in a horizontal section vs. conveyor belt impact on cleaning

Cuttings accumulation tends to form along the low side of the wellbore when the BHA remains static without rotation or circulation. This can lead to increased torque, drag, or risk of pack-off, particularly in extended-reach wells or lower-inclination zones.

With the introduction of Artificial Intelligence (AI) & dynamic surveying, bandwidth availability increased significantly, enabling real-time high-resolution image transmission at ROPs more than double (~130% higher) the previous limitations in similar environments. The objective in this campaign was to move beyond traditional telemetry constraints by integrating intelligent telemetry, AI-enhanced transmission, and real-time operational decision-making. By combining continuous downhole measurements with dynamic connection workflows, the campaign demonstrated that significant performance gains were achievable without compromising safety, borehole quality, or image resolution.

Engineering Workflow and Methodology

The engineering workflow implemented centered on enabling high-performance drilling without compromising data quality. The foundation of this transformation was the deployment of an AI-powered high-speed telemetry system that allowed real-time transmission of Deep Resistivity and Ultra-deep Resistivity images at high ROP. This real-time imaging capability enhanced geosteering decisions by

providing immediate visual feedback during horizontal drilling, reducing the reliance on post-run memory-mode data.

To fully leverage this capability, the team integrated a connection optimization strategy through dynamic surveying, enabling continuous trajectory updates while minimizing the time spent stationary in the wellbore. Real-time monitoring of T&D (torque and drag), standpipe pressure (SPP), and equivalent circulating density (ECD) enabled safer, faster connection exits without compromising borehole integrity.

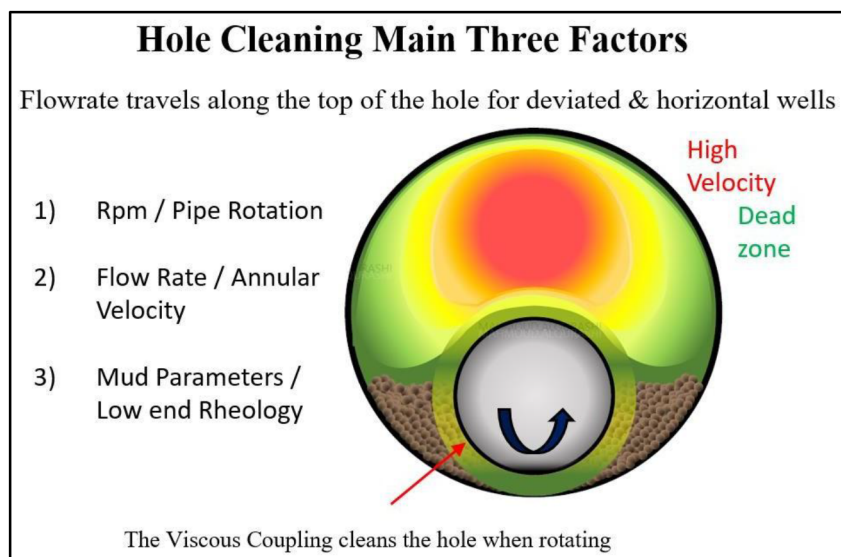


Figure 2—Hole cleaning mechanism in deviated and horizontal wells

Efficient hole cleaning relies on three interdependent factors: pipe rotation (RPM), annular velocity (flow rate), and low-end rheology of the drilling fluid. The diagram shows fluid traveling along the top side of the borehole, while cuttings tend to settle in the low-side dead zone. Pipe rotation generates viscous coupling, mobilizing cuttings and enhancing low-side transport. This combined approach supported a step-change in efficiency and real-time decision-making. By aligning telemetry performance with operational workflows, the team was able to enhance drilling consistency, maintain image resolution at high ROPs, and reduce time spent on surveys and connections — all without compromising data quality or operational safety.

AI-Driven Real-Time Telemetry

It was a coupled solution of an AI-enhanced telemetry system along with the dynamic surveying capabilities. It uses adaptive data compression and intelligent prioritization to transmit high-resolution LWD images — including deep and ultradeep resistivity — in real time, even under high-ROP and dynamic downhole conditions. Real-time image data was already being used for geosteering. However, to obtain clear and reliable images, operations were typically accustomed to lower ROPs — to minimize noise and avoid image degradation. This introduced a trade-off between drilling efficiency and image quality, limiting the ability to optimize both simultaneously.

The introduction of high-speed intelligent telemetry changed this balance. By enhancing telemetry performance and improving real-time image clarity at the surface, system enabled geoscientists to make more accurate geosteering decisions, more quickly, and with greater confidence. Image quality was not only preserved — it was significantly improved — even as on-bottom ROPs were pushed up to 3X times. This new capability allowed the team to safely increase drilling speed without compromising visibility or placement precision.

Survey Time Elimination using Dynamic Surveying

System enabled continuous directional measurements while rotating, eliminating the need to halt drilling for static surveys. Survey data were captured seamlessly during pipe connections, reducing the time the BHA remained stationary in open hole. Minimizing stationary time helped mitigate risks such as cuttings buildup, elevated torque and drag, or pack-off — especially in reactive or poorly consolidated formations. By enhancing real-time data transmission, system improved situational awareness for directional control and trajectory decisions. Combined with the AI powered telemetry platform, this coupled solution ensured continuous delivery of high- resolution resistivity images even at high ROPs. This capability supported faster, more confident geosteering decisions without relying on memory-mode dumps or slowing down for telemetry limitations.

Connection Practice Optimization

Connection procedures were re-engineered using real-time monitoring of torque and drag (T&D), standpipe pressure (SPP), and equivalent circulating density (ECD). When these parameters remained stable—backreaming was omitted. This risk-based connection strategy significantly reduced connection times without compromising hole quality or introducing additional mechanical risks.

Together, these systems created a unified performance framework that minimized delays between stands, preserved borehole integrity, and enabled fast, precise drilling across extended lateral sections.

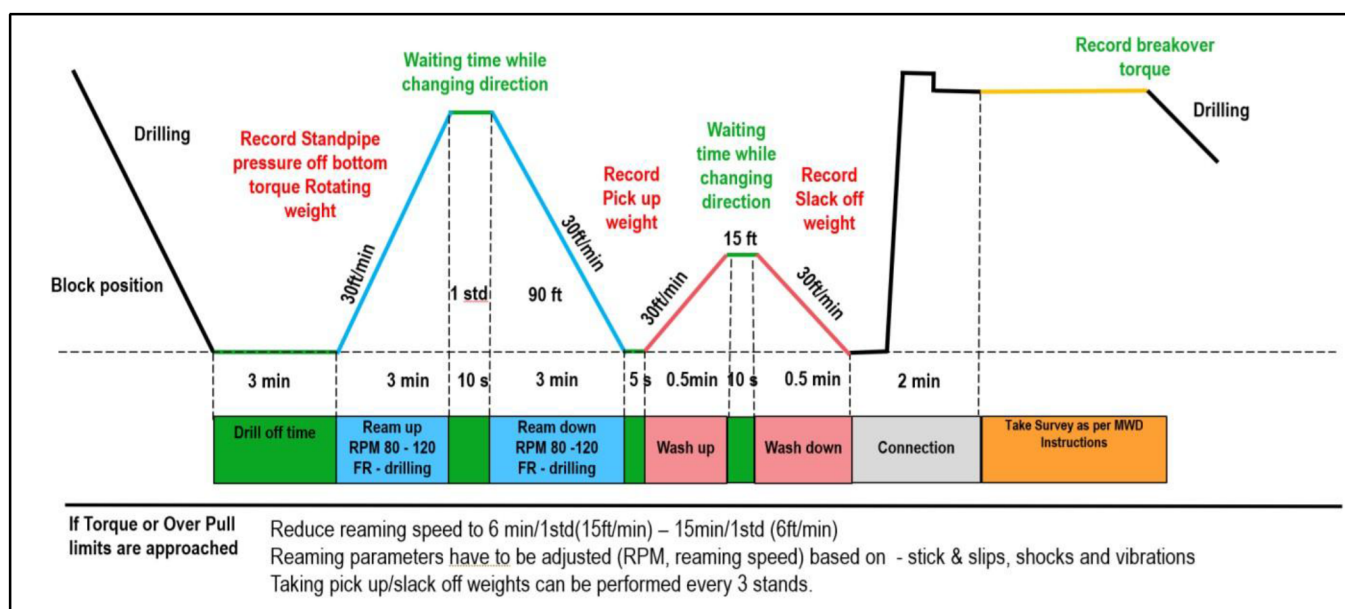


Figure 3—Standard Connection Diagram for 8-1/2" Horizontal section

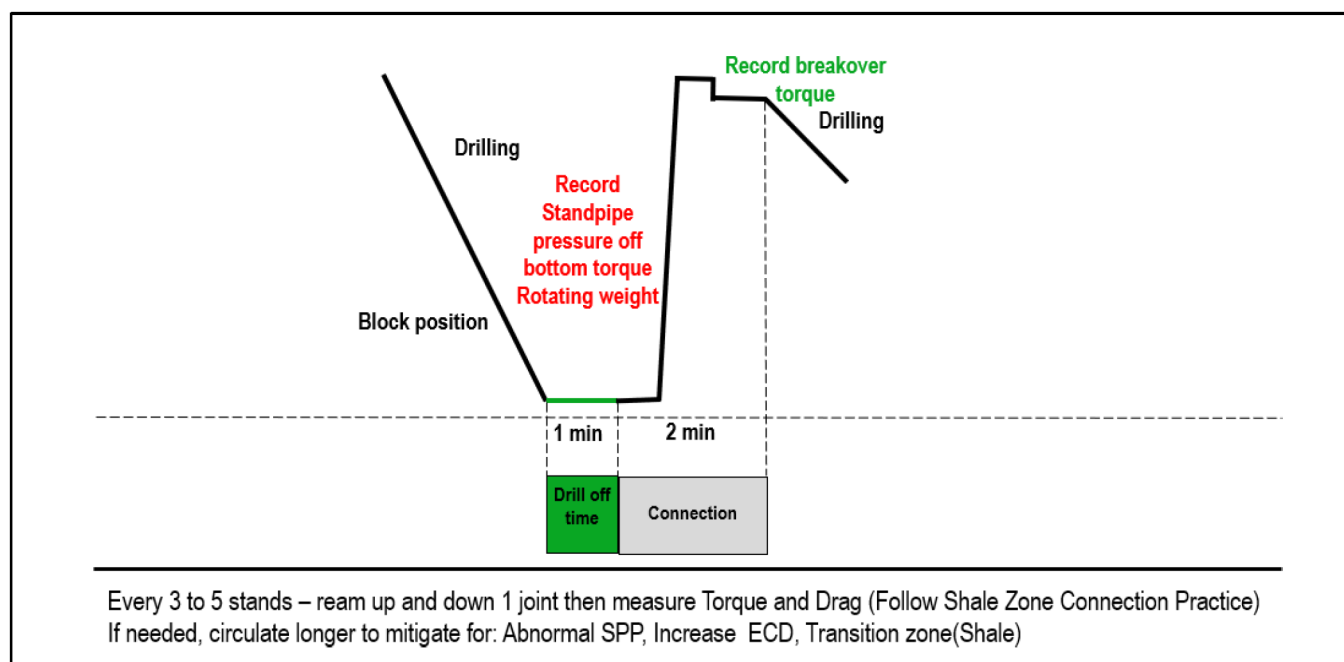


Figure 4—Fast Connection Practice breakdown - Sand Zone

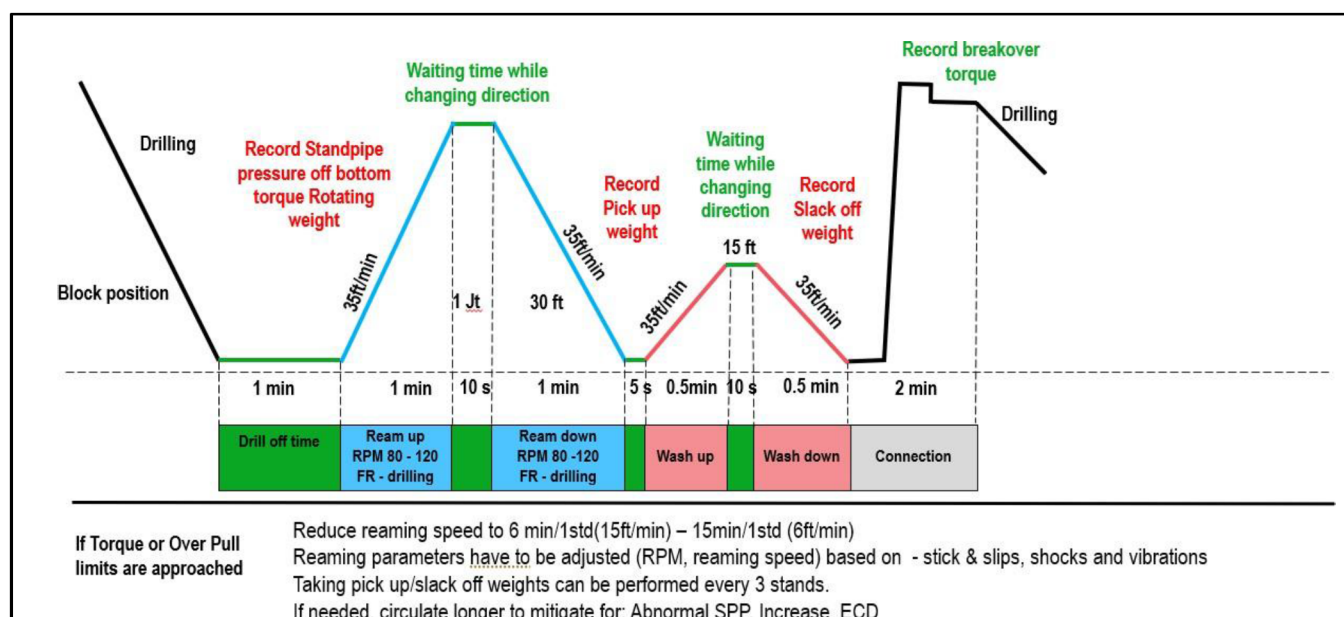


Figure 5—Fast Connection Practice breakdown - Shale Zone

Results

The dynamic surveying capability eliminated the need to pause drilling for static surveys after each connection. By providing continuous directional data, system enabled immediate drill-ahead following pipe makeup / connection. This allowed a smoother operational rhythm across multiple wells. In addition, avoiding prolonged stationary time at connection depth helped reduce the risk of stuck pipe, particularly in unstable hole sections.

The integrated deployment of AI enhanced telemetry, dynamic surveying and real-time engineering workflows delivered consistent performance improvements across multiple wells.

Drilling Performance

The average rate of penetration (ROP) in the 8.5-in lateral sections increased by ~130% compared to field average baseline.

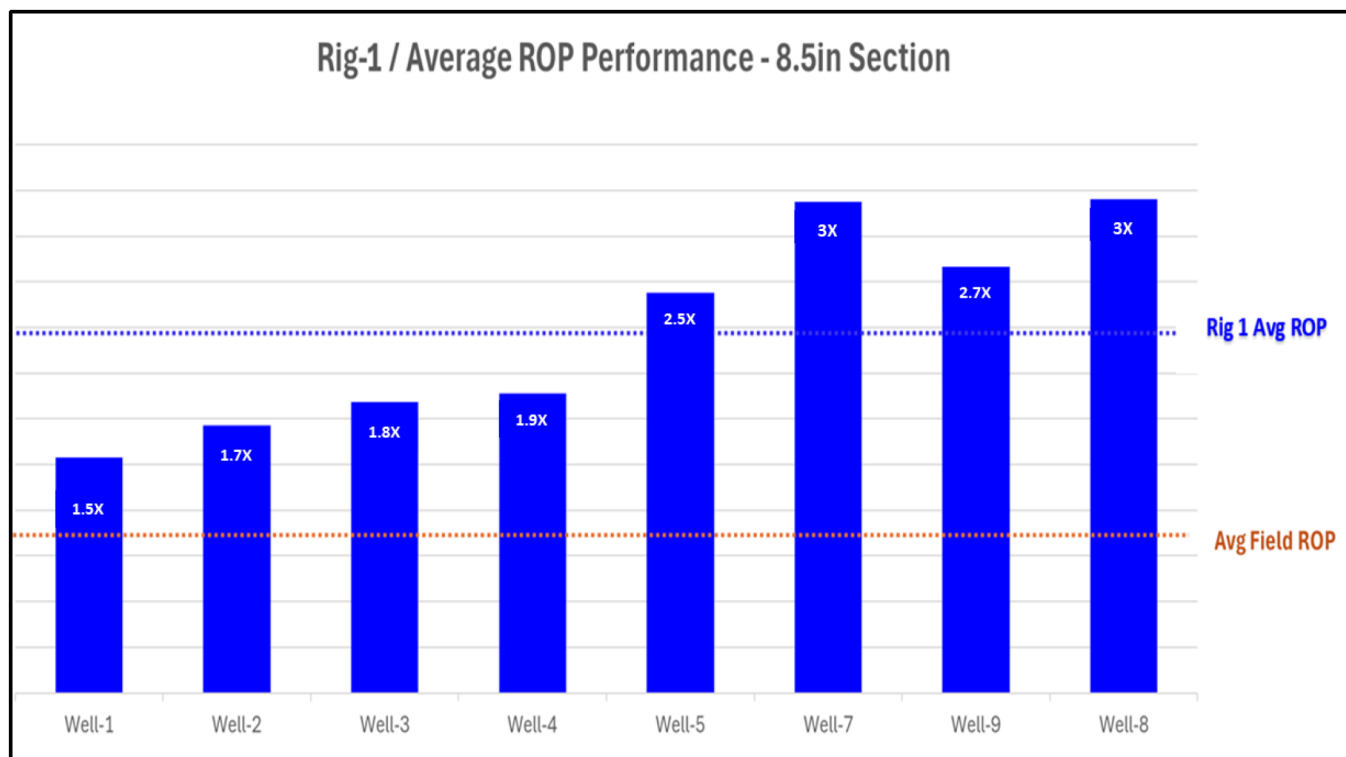


Figure 6—Average ROP in the 8.5-in section showing 130% improvement

Importantly, this level of performance was not limited to a single well. The improvements were repeated and further optimized across eight consecutive wells, demonstrating consistent reliability and reinforcing the scalability of the approach. Each new well applied the same workflow with incremental refinements, confirming operational robustness under varying formation and rig conditions.

Connection Time Savings

Connection time was reduced significantly across the campaign. Prior to implementation, durations typically ranged between 23 to 28 minutes. In the Rig-1 wells, connection times ranged from 8.5 to 14.5 minutes, with an average of 12.31 minutes.

In Well-7, of all connections, only five instances required backreaming — none exceeding 30 ft — demonstrating that real-time mechanical stability monitoring enabled efficient connections with minimal need for pullback, contributing directly to time savings.

Rig-1 connection efficiency steadily improved through a staged deployment of revised practices tailored to downhole conditions. Initially, in the first five wells (Well-1 to Well-5), a unified fast connection approach was implemented (Figure 5), replacing the more conservative procedure (Figure 3). This method, though not formation-specific at the time, already reduced average connection times from over 14 minutes to approximately 13 minutes.

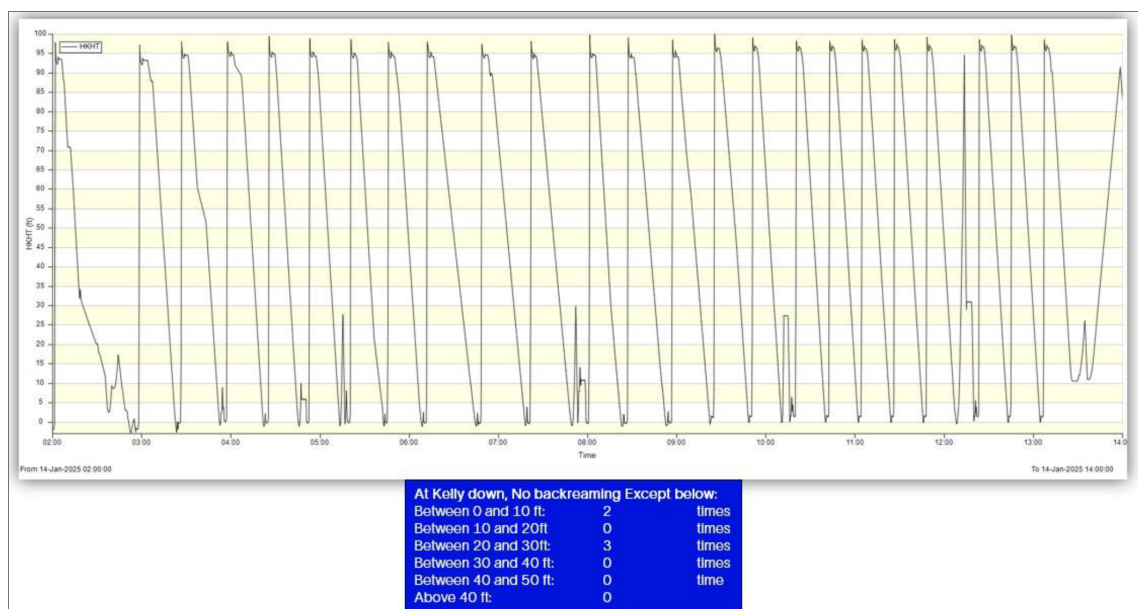


Figure 7—Hookload profile and backreaming depth per connection in the 8.5" section of Well-07

As operational understanding matured, it became evident that additional optimization was achievable in clean formations. A new formation-dependent strategy was adopted: the Fast Connection Practice for Sand Zone (Figure 4), which eliminated backreaming and further reduced connection times to below 10 minutes. The final three wells (Well-7 to Well-9), shown in purple in Figure 8, combined both shale- and sand-specific connection procedures. This adaptive methodology ensured both time savings and hole condition integrity across varying lithologies.

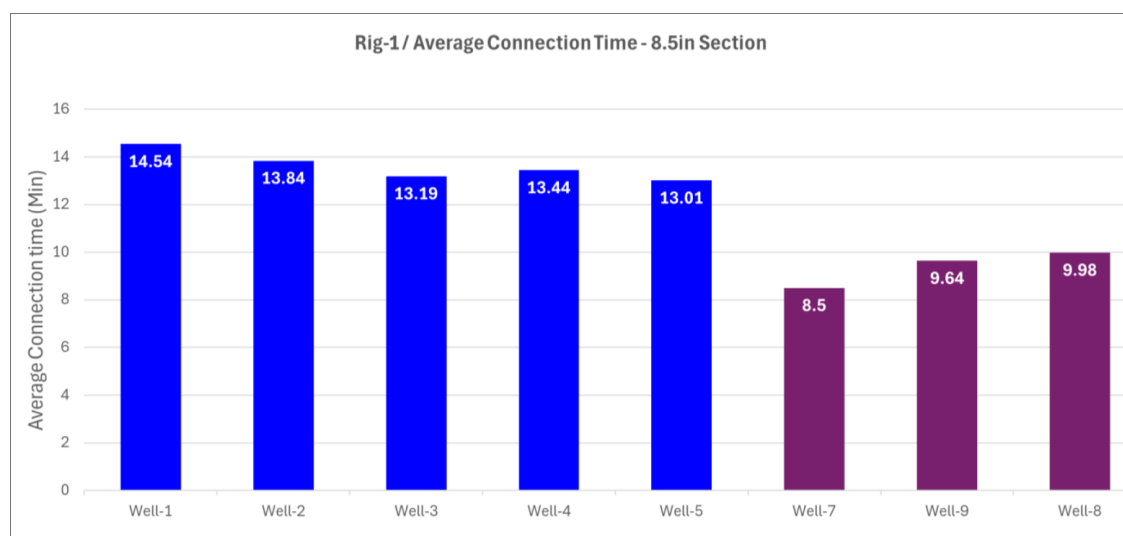


Figure 8—Connection time per well showing the impact of optimized connection strategy

With an average of 33 connections per well in the 8.5-in lateral section, this reduction translated to a saving of 13.2 minutes per connection, or approximately 7.3 hours saved per well from connection time optimization alone.

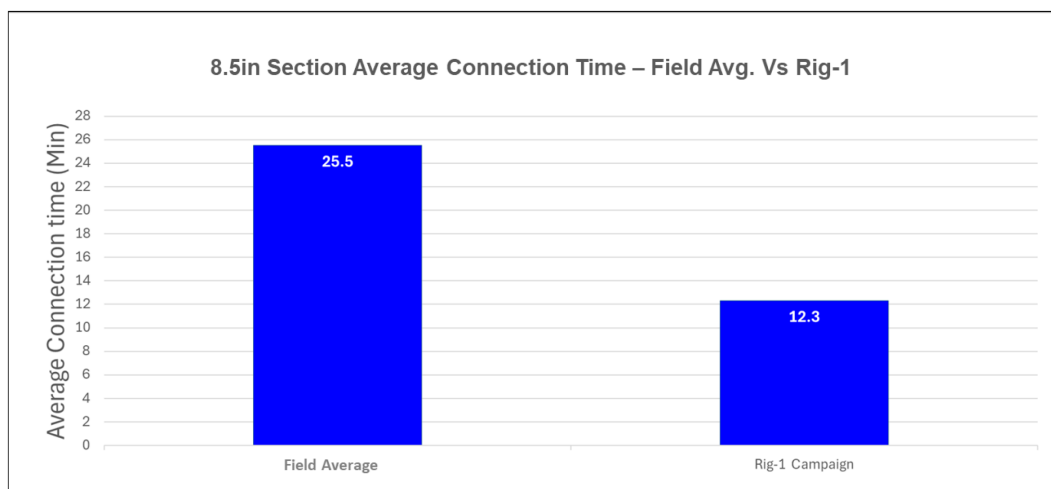


Figure 9—Average Connection Time Reduction Before and After Workflow Implementation

Total Rig Time Reduction

Drilling hours for the 8.5-in lateral section were reduced from a previous average of 40 hours to 19 hours in the best-performing wells. This corresponds to a rig time saving of 21 hours per well in the lateral interval alone.

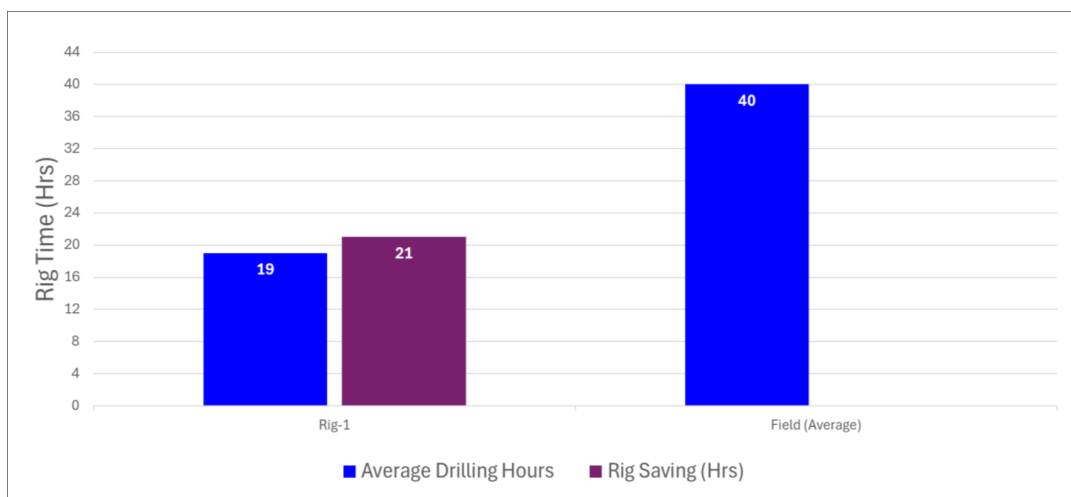


Figure 10—Total rig time comparison between field average and Rig-1 wells

Data Quality and Operational Continuity

High-resolution real-time images were successfully acquired and interpreted at full drilling speed, supporting continuous geosteering without interruption. No drilling dysfunctions, non-productive time (NPT), or image transmission failures were reported during the campaign, confirming the reliability, repeatability, and robustness of the workflow. Following the success of this implementation, the same combined approach is now being adopted in other rigs, and fields, reinforcing its broader applicability.

Hole Cleaning and ECD Management

As drilling performance improved and ROP increased up to 130%, real-time monitoring of equivalent circulating density (ECD), torque, and drag was critical to maintaining wellbore integrity and effective hole cleaning. This continuous surveillance enabled proactive adjustments when needed, allowing high ROP to

be sustained without compromising hole conditions. The strategy was not to limit ROP arbitrarily, but to control ROP only when required, particularly in transition zones characterized by shaly sand formations, to ensure stable parameters and uninterrupted progress.

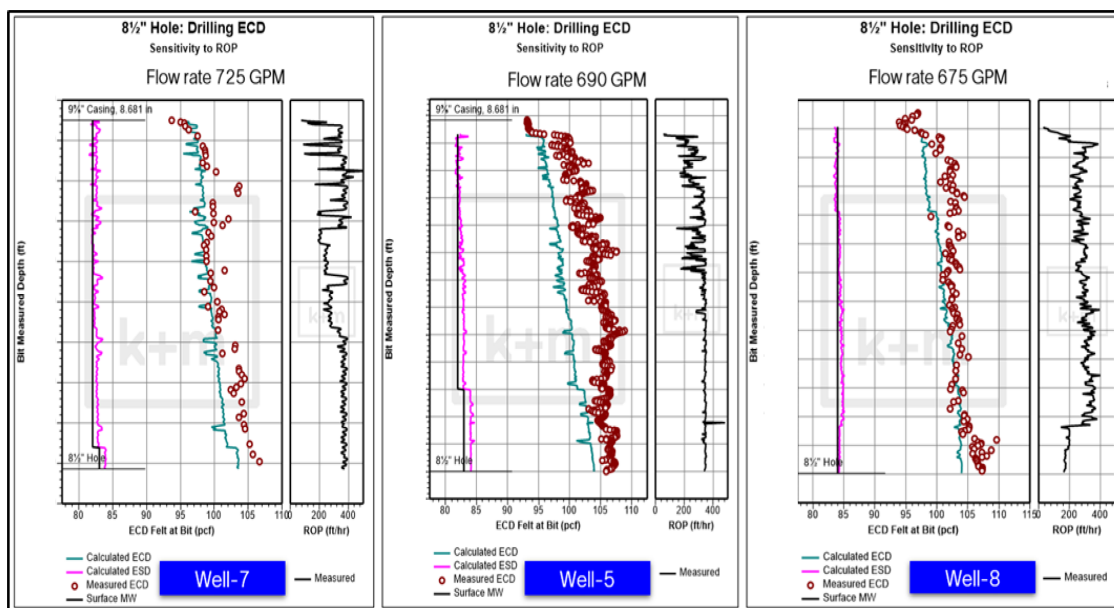


Figure 11—Measured and simulated ECD trends across Wells 5, 7, and 8

Higher ROPs and varying flow rates influenced ECD behavior across all wells. Real-time monitoring of ECD supported optimized drilling without compromising hole conditions, especially in transition zones characterized by shaly sand intervals.

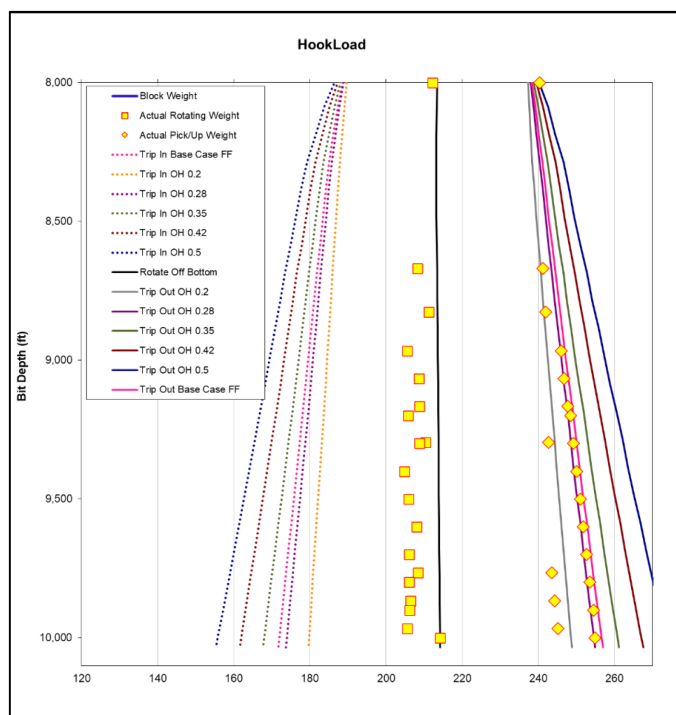


Figure 12—Modeled vs. Actual Hookloads for Torque & Drag validation

In addition to ECD monitoring, Torque and Drag (T&D) models were used continuously to validate hole condition and detect early signs of cuttings accumulation or borehole instability. Actual hookload values were compared against simulated curves during tripping and rotation to ensure alignment with expected friction factors. This provided a second layer of assurance that the hole remained clean and mechanically stable, enabling high ROP to be sustained without introducing additional risk.

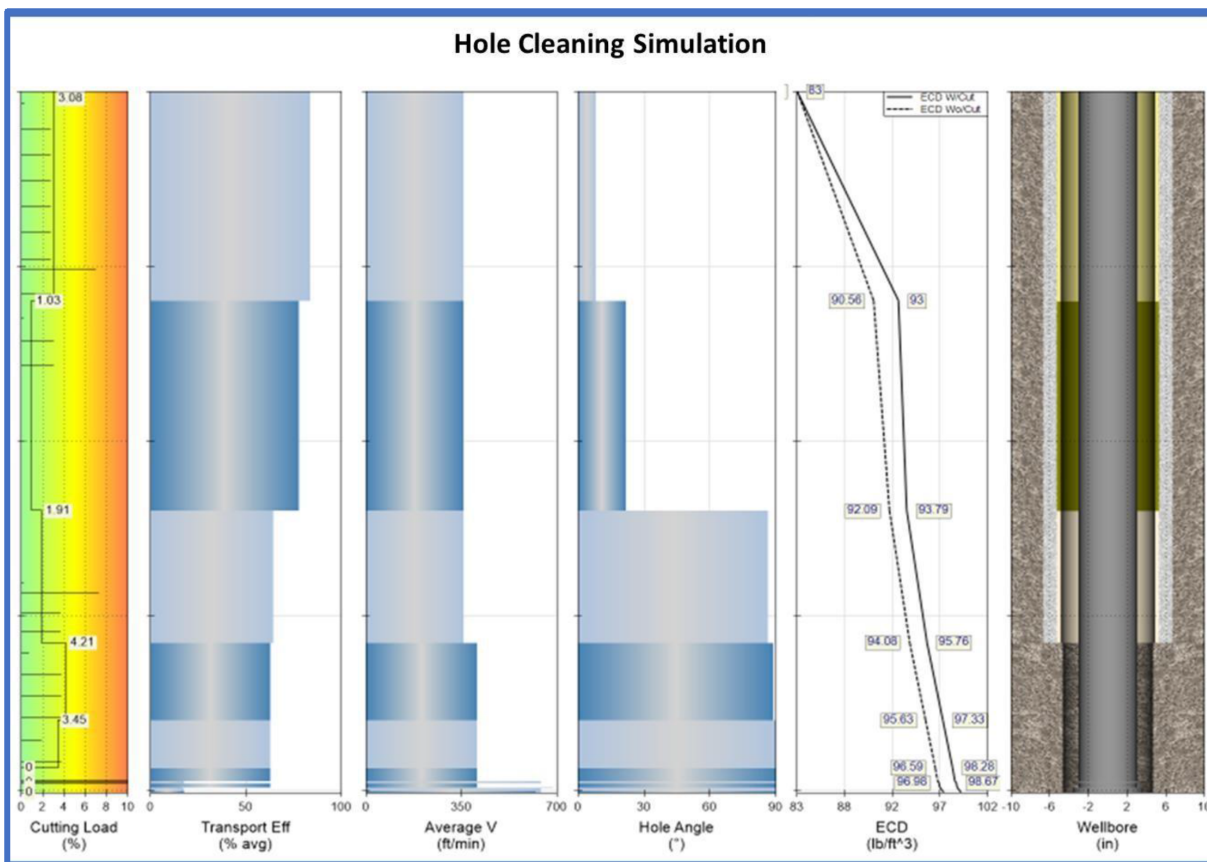


Figure 13—Hole cleaning & ECD management

While drilling sand, the total annular cuttings load was maintained at approximately 2.37%, based on an assumed cuttings diameter of 0.1–0.2 in. Localized zones with cuttings loads approaching 3% were managed without any significant operational impact.

In contrast, while drilling shale or shaly sand intervals, cuttings are expected to be larger—typically ranging from 0.2 to 0.35 in or more—depending on borehole size and the orientation of horizontal formation stresses. In these intervals, ROP should be evaluated carefully in real time based on cuttings shape, size, and volume observed at the shale shakers to avoid accumulation risks and maintain cleaning efficiency.

Rheological parameters remained within optimal ranges, supporting solids suspension and transport. Gel strengths and fluid loss control were consistent across wells, ensuring borehole stability. Low PPT values indicated strong filter cake performance, contributing to minimized formation invasion and improved wellbore integrity.



Figure 14—Summary of drilling fluid properties across selected wells: PV/Yp, gel strengths, HTHP filtrate loss, and PPT

Discussion

The integration of AI-driven telemetry, dynamic surveying, and real-time engineering workflows demonstrated that drilling efficiency can be significantly improved without compromising data quality or operational safety. A key breakthrough was the ability to maintain high-resolution LWD imaging while drilling at ROPs up to 3X times. Although geosteering has always relied on real-time data, the typical limitation has been the lower quality of real-time images compared to recorded-mode data, which is only accessible after drilling is complete. In this campaign, the ability to access high-quality images in real time provided geologists with enhanced visibility and confidence in well placement, allowing more accurate lateral navigation and reducing the risk of geosteering uncertainty.

Connection time optimization further contributed to performance gains. By closely monitoring torque and drag (T&D), standpipe pressure (SPP), and equivalent circulating density (ECD) during each stand, the team was able to skip backreaming when hole conditions remained stable. This not only reduced connection durations but also improved overall drilling rhythm without introducing mechanical risk.

The elimination of static surveys using dynamic surveying practices also played a major role. Continuous directional measurements allowed for immediate drill-ahead after each connection, avoiding extended stationary periods that are typically required for survey acquisition. In addition to saving time, this reduced exposure to risks such as stuck pipe or pack-off, especially in mechanically sensitive intervals.

These improvements were not limited to a single well. The workflow was successfully repeated and refined across eight consecutive wells, proving its consistency and robustness under varied field conditions. As a result, the approach is now being deployed in other rigs, and fields, reinforcing its value as a scalable and field-proven drilling performance solution.

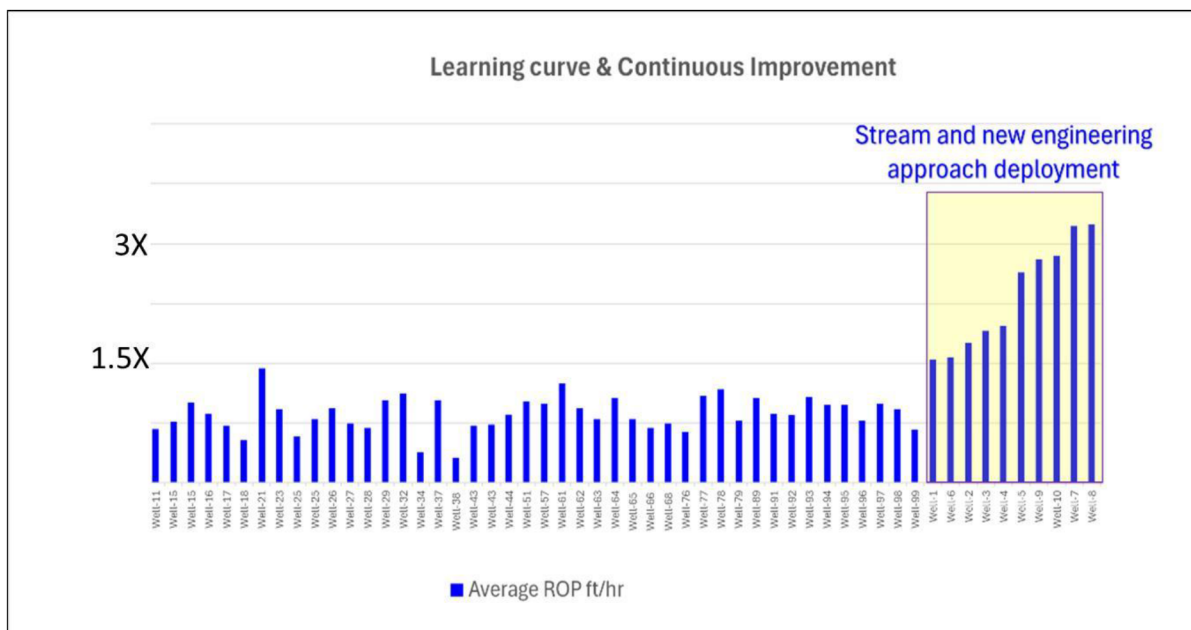


Figure 15—Learning curve showing performance improvement and consistency

Conclusion

The recent drilling campaign demonstrated that the integration of AI-enhanced telemetry, continuous dynamic surveying, and real-time connection optimization can significantly improve drilling performance without compromising borehole quality or data resolution. Average ROP more than doubled compared to the Field baseline, while connection time and static survey delays were substantially reduced. These improvements translated to a total rig time saving of 21 hours per well in the 8.5-in lateral section.

One key breakthrough was the removal of ROP as a limiting factor for real-time image transmission. During the campaign, high-resolution images were reliably transmitted at ROPs up to 3X times—yet no new technical limiter was observed. With increased confidence in hole condition monitoring and telemetry stability, the team now has the operational envelope to explore even higher ROPs.

Another lesson from the campaign was the importance of moving beyond standard conservative routines—such as default backreaming after every connection—toward condition-based decisions supported by real-time indicators. When torque and drag (T&D), standpipe pressure (SPP), and ECD remained stable, backreaming was safely eliminated, leading to meaningful time savings without increasing risk.

The workflow was applied consistently across eight wells, confirming its robustness and repeatability. Following its success, the approach is now being adopted in other wells, validating its scalability and long-term operational value.